

COMB-05 - Teacher Diagnostic Key

Recognition / First-Line Lab

1. Boxes are residues modulo 16.
2. Ten club boxes; if every box had at most 3, total capacity would be 30.
3. Divide the square into four $1/2 \times 1/2$ squares; each has diameter $\sqrt{2}/2$.
4. Pair boxes $\{1,10\},\{2,9\},\{3,8\},\{4,7\},\{5,6\}$.
5. Select an element that is extremal for the property used by the closure operation; the extremal choice must forbid a strictly better admissible element.
6. Choose a shortest interval.
7. Inclusion-exclusion/complement counting: the task asks for an exact count with overlapping forbidden properties.
8. Inequality optimization: the domain is continuous and the target is a numerical extremum.
9. Twelve residue boxes modulo 12; two coincide.
10. Partition the triangle into 9 congruent equilateral triangles of side $1/3$; 10 points force a shared small triangle, whose diameter is $1/3$.
11. Pair $\{1,2\},\{3,4\},\dots,\{99,100\}$; 51 choices force a full pair.
12. Pick one person P. Among the other 5, at least 3 are all acquaintances of P or all strangers to P. In the first case, if any two of those 3 know each other there is an acquaintance triangle; otherwise those 3 form a stranger triangle. The second case is complementary.

Practice / Transfer

1. $\text{ceil}(25/6)=5$.
2. Seven residue classes modulo 7.
3. One hundred residue classes modulo 100.
4. $\text{ceil}(40/12)=4$.
5. Ten boxes $\{1,2\},\{3,4\},\dots,\{19,20\}$.
6. Four half-size squares.
7. Nine congruent subtriangles of side $1/3$.
8. Six pair boxes summing to 13.
9. Four residue classes modulo 4.
10. The stated consequence is exactly the defining no-improvement property of the closest pair.
11. $\deg(v_{\max}) \geq \deg(v)$ for every vertex v .
12. A strictly smaller nonempty member of the family would contradict the choice.
13. Same pairing proof as item 5, with 50 adjacent pairs. Enumeration does not contribute because only forced existence is requested.
14. Partition the side-2 square into 16 squares of side $1/2$; 25 points force two in one cell, distance at most $\sqrt{2}/2$.
15. Same proof as Recognition item 12.
16. There are $C(20,4)=4845$ increasing quadruples. Direct counting of $a+c=b+d$ gives 525 balanced, hence 4320 unbalanced. Among 4411 selected, at least $4411-4320=91$ are balanced.
17. An average forces an inequality threshold, not an exact occupancy.
18. Twelve residue classes solve the existence claim immediately; exact overlap counts are unnecessary.
19. The structure is discrete/local; the chosen edge must be used to forbid a smaller admissible edge or force a structural contradiction.
20. Same-cell membership is useful only if it implies the desired distance bound; diameter, not area, controls that implication.

Mastery

1. residues modulo 19.
2. $\text{ceil}(37/9)=5$.
3. Seven boxes $\{1,14\},\dots,\{7,8\}$.
4. four half-size squares.
5. fifty adjacent pairs.
6. Ramsey $R(3,3)=6$ elementary proof as above.
7. Choosing an extreme object is only useful when the operation or constraints would produce a strictly smaller/larger admissible object, contradicting extremality. No such consequence has been supplied.
8. Pigeonhole establishes existence/capacity, not the exact union size with overlaps.
9. This is a continuous numerical bound; use an inequality/equivalent algebraic representation.
10. 525. A direct trace: for each $b < d$, let $s = b + d$; count $a < b$ with $c = s - a > d$ and $c \leq 20$. Summing gives 525.
11. 91.
12. Partition the side-2 square into four unit squares; 17 points force at least 5 in one unit square, but that alone is not enough. Instead partition into 16 half-unit squares; 17 points force two in one, giving distance at most $\sqrt{2}/2$, which is stronger than $\sqrt{2}$.

Historical source audit

- IOQM-2023-Q18: verified answer 71. Teacher-side derivation: add all 50 polygon sides to the selected diagonals. The selected-diagonal condition gives an outer 1-planar graph; the tight edge bound is $5n/2 - 4 = 121$, so at most 71 diagonals. A fan of 47 diagonals from one vertex plus 24 alternating short diagonals attains 71. This density lemma is source-analysis material, not a Grade-9 prerequisite doctrine.
- IOQM-2023-Q27: verified answer 91 by the 525-balanced / 4320-unbalanced count above.

All learner items and answers were independently recomputed before freezing.

COMB-05 - Source Coverage Map

Authority

Canonical historical IDs follow the frozen 90-question corpus. This topic owns pigeonhole and extremal selection; geometry and number theory are transfer surfaces, not alternate canonical owners.

Stable ID	Source/key status	Verified answer	Role	Source-integrity status
IOQM-2023-Q18	HBCSE-linked MTAI paper with embedded key	71	primary extremal/geometry anchor	
CLEAN_VALIDATED				
IOQM-2023-Q27	HBCSE-linked MTAI paper with embedded key	91	primary extremal-complement anchor	
CLEAN_VALIDATED |

Validated paper/key: https://www.mtai.org.in/wp-content/uploads/2023/09/IOQM_Sep_2023_Question-paper-with-answer-key.pdf.

The metadata-correction overlay does not modify either anchor.

Independent mathematical traces

IOQM-2023-Q18

The source surface is a convex 50-gon with a selected family of diagonals under the local rule that each selected diagonal intersects at most one other selected diagonal. Add the 50 polygon sides. The resulting drawing is an outer 1-planar graph, whose tight edge bound is $5n/2-4$. With $n=50$, at most 121 total edges occur, hence at most $121-50=71$ selected diagonals. A construction with 47 fan diagonals from one vertex plus 24 alternating short diagonals attains 71. Independent result: 71, agreeing with the validated key.

The outer-one-crossing density lemma belongs in teacher/source analysis; the learner-facing doctrine is the extremal translation from a local crossing cap to a global density restriction.

IOQM-2023-Q27

Each increasing 4-subset of $\{1, \dots, 20\}$ is uniquely written $a < b < d < c$. Call it balanced when $a+c=b+d$. There are $C(20,4)=4845$ increasing quadruples. Direct summation over the middle pair (b,d) gives 525 balanced quadruples, so there are 4320 unbalanced. A selected family of 4411 can contain all 4320 unbalanced quadruples, but every additional member must be balanced. Hence at least $4411-4320=91$ balanced quadruples are forced. Independent result: 91, agreeing with the validated key.

Coverage implications

Historical recurrence is not treated as official weightage. The two anchors support different mechanisms: local-to-global extremal density and extremal complement/capacity. Direct/generalized pigeonhole, geometric cells, residue classes, nearest/farthest choices, and close method contrasts are completed with author-created items whose provenance is explicit in Item_Metadata.csv.